

ARCHITECT REVIEW

SALDEV 1427 QCP Cloning Logic Architect Review

FlywirePartial | Read-only stepped-pricing JavaScript and Apex best-practice review

Source	SALDEV-1427 FlywirePartial Enterprise sandbox 00DhG0000000jOXUAY
Scope	2 QCP scripts 1 trigger 1 handler 1 test class
Review date	4 September 2026
Validation mode	Live read-back JS syntax focused Apex tests check-only
Decision	NO-GO - Changes Requested before QA or promotion
Boundary	Read-only review; no record, metadata, activation, or deployment changes

ARCHITECT DECISION

Changes Requested. SALDEV-1427 is not release-ready: QCP dependencies are incomplete, Domestic_Sponsor__c is not a real quote-line field, usage fields are simultaneously read-only and editable, lineage matching is ambiguous, and check-only failed at 43.59% trigger coverage.

Acceptance and quality outcome

#	Scope	Result	Evidence
1	ARR values cleared on SUP clones	NOT PROVEN	Nulling lists exist, but dependencies omit the fields and one API name is invalid.
2	First group remains editable	PARTIAL	An editable allowlist exists; no live QLE functional run was performed.
3	Later groups are restricted	DEFECT	The script reads group Source and Number, but they are not declared dependencies.
4	Usage price fields editable	FAIL	Usage_Rate__c and Usage_Rate_Amt__c are also globally read-only; that rule wins.
5	Opportunity Line ID lineage	HIGH RISK	Quote + Product cannot distinguish group, bundle, configuration, or ship-to context.
6	Required scenario matrix	FAIL	No JavaScript unit evidence; Apex assertions miss Group 2; check-only failed coverage.

Runtime path and defect concentration

SALDEV-1427 spans QCP configuration, QLE editability, and Apex Opportunity Line ID lineage. The code has a useful bulk-processing base, but several runtime contracts and the release gate remain incomplete.

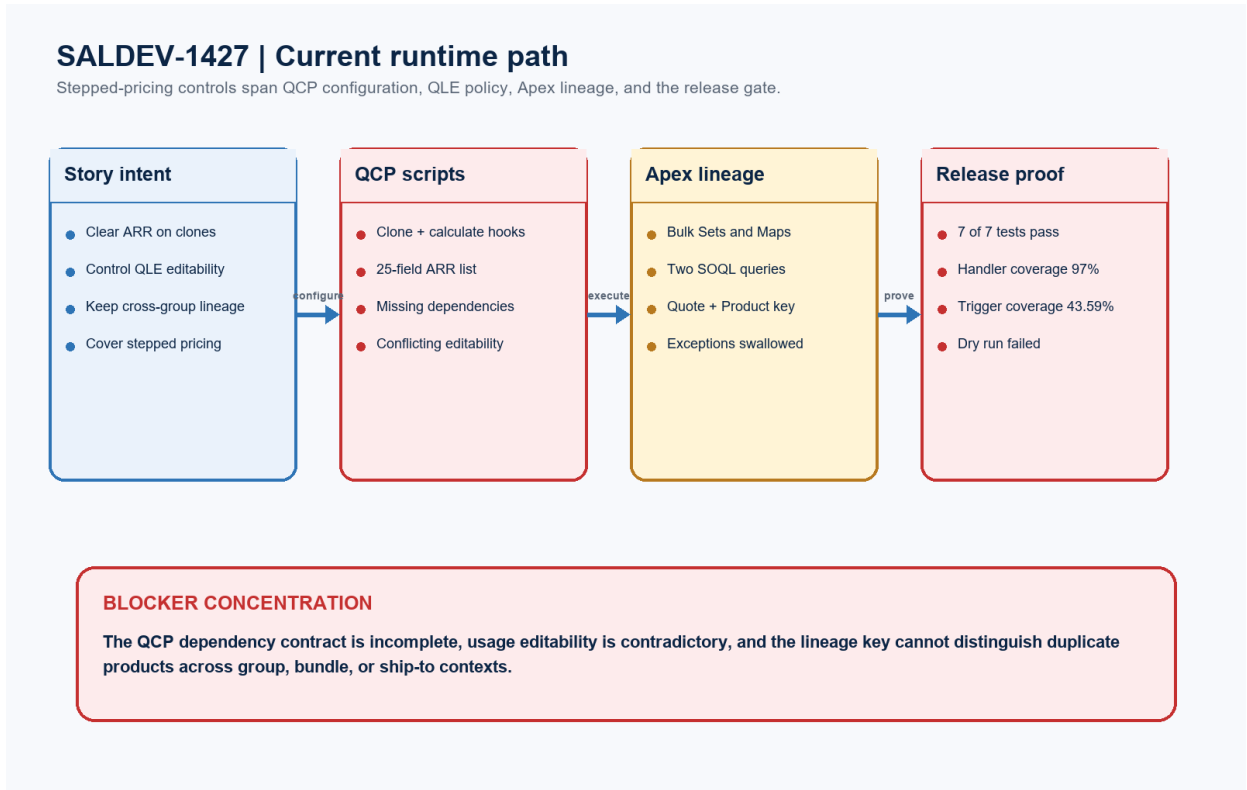


Figure 1. SALDEV-1427 implementation path and release blockers

DESIGN READING

Blue shows the intended contract. Green shows practices worth retaining. Amber marks incomplete proof. Red identifies issues that can create incorrect quote-line behavior or block release.

Detailed code and validation evidence

1. Live component and runtime evidence

Evidence area	Observed result	Evidence
Target org	Verified sandbox	FlywirePartial is org 00DhG0000000jOXUAY; Enterprise Edition; IsSandbox=true.
QCP_HideFieldIn QLE	Runtime present	1,999 JavaScript lines; transpiled runtime exists; syntax check passed.
QLE_CloneLineHandler	Runtime present	58 JavaScript lines; clone hook and runtime exist; syntax check passed.
Apex runtime	Bulk-structured	Trigger delegates to a with-sharing handler using Sets, Maps, and two SOQL queries outside loops.
Field read-back	24 of 25 found	Domestic_Sponsor__c does not exist; Processed_Volume_Amount__c is correctly excluded.

2. Best-practice findings

Finding	Severity	Evidence
Dependency contract	BLOCKER	Clone script omits all ARR dependencies; QCP omits required line and group dependencies.
Rule precedence	BLOCKER	Two usage fields are both globally read-only and ramp-editable; read-only executes first.
Lineage determinism	HIGH	Occurrence order can cross bundles or ship-to groups; fallback reuses the first master ID.

3. Test and validation evidence

Test / boundary	Status	Evidence
Focused Apex suite	PASS	7 of 7 QuoteLineTriggerHandlerTest methods passed; zero failures.
Handler coverage	PASS	QuoteLineTriggerHandler measured 97%.
Trigger coverage	FAIL	QuoteLineTrigger measured 43.59%; Salesforce requires at least 75%.
Check-only package	FAILED	0AfhG000001da33SAA: 3/3 components and 7/7 tests, but coverage failed; nothing deployed.
Behavior proof	MISSING	No JavaScript suite or live QLE matrix; the second-group Apex outcome is not asserted.

Required target architecture and proof model

The release-ready design should centralize the ARR field contract, declare every QCP dependency, make editability precedence explicit, correlate lineage deterministically, and prove JavaScript and Apex behavior independently.

SALDEV-1427 | Required release design

A canonical field contract, ordered policy, deterministic lineage key, and layered tests

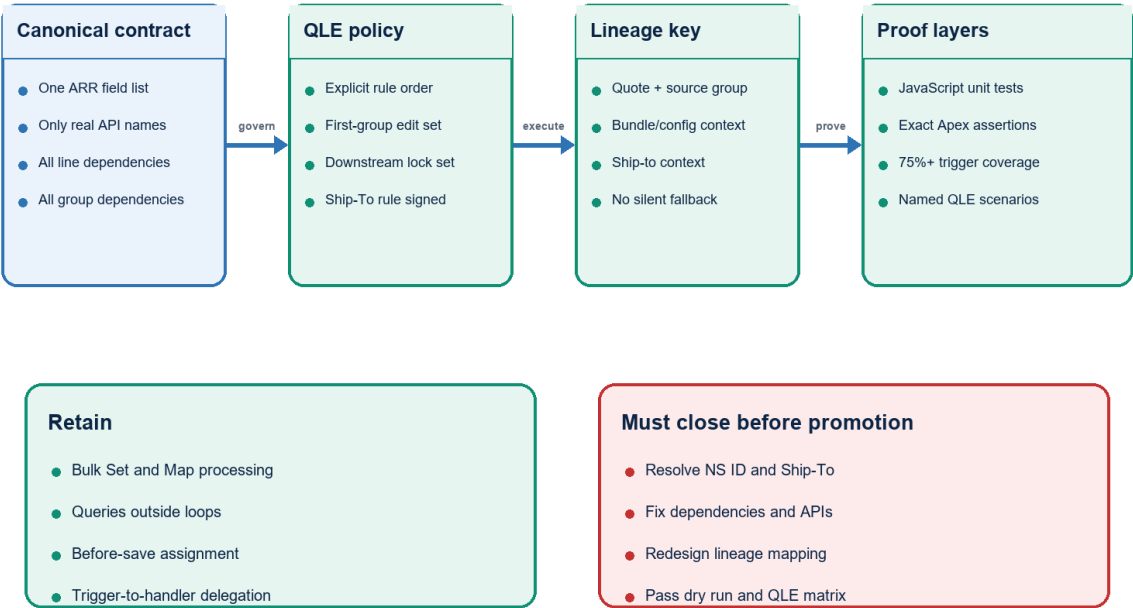


Figure 2. SALDEV-1427 required data contract, lineage design, and proof layers

Architect interpretation: the Apex handler is bulk-structured, but QCP configuration and identity semantics are executable parts of the solution. Passing handler tests cannot compensate for missing dependencies, contradictory rules, ambiguous lineage, or insufficient trigger coverage.

Required completion gates

Gate	Required action	Owner	Evidence needed
1	Correct APIs and QCP dependencies	CPQ Developer	All line/group APIs exist and are declared; remove or correctly map Domestic_Sponsor__c.
2	Make editability precedence explicit	CPQ Developer	Usage fields work as intended in first and downstream groups.
3	Redesign lineage correlation	Apex Developer	Deterministic source/group/bundle/ship-to key; no silent fallback.
4	Resolve story semantics	Product Owner	Signed NS ID, Ship-To, amendment, and SALDEV-1404 dependency decisions.
5	Add exact automated tests	Developer / QA	JavaScript tests plus Apex assertions for every group, duplicate product, bundle, and ship-to case.
6	Re-run release and QLE proof	Dev Lead / QA	75%+ trigger coverage, successful check-only, and named SUP functional read-back.

Ready-to-paste architect response

STORY COMMENT

Architect review completed for SALDEV-1427 in FlywirePartial. Decision: Changes Requested / NO-GO before QA or promotion. Both QCP scripts pass syntax checks; the with-sharing Apex handler is bulk-structured; 7 of 7 focused tests passed; and handler coverage is 97%. Blockers remain: incomplete QCP dependencies, nonexistent Domestic_Sponsor__c, conflicting usage-field editability, ambiguous Quote + Product lineage, swallowed exceptions, and missing JavaScript and exact Group 2 assertions. Check-only 0AfhG000001da33SAA validated 3 of 3 components and ran 7 passing tests, but failed because QuoteLineTrigger coverage is 43.59%, below 75%. Nothing deployed. Resolve requirements, correct the field contract, redesign deterministic mapping, add the missing tests, pass check-only, and complete the named stepped-pricing QLE matrix before resubmission.

Evidence boundary

Evidence: the SALDEV-1427 source document, verified FlywirePartial sandbox and scripts, retrieved Apex and field metadata, syntax checks, focused tests, and check-only validation. No Salesforce record or metadata changes, deployment, activation, promotion, or Production action occurred.